Register Number: 192425119

Question 5: Traffic Density and Accident Severity

Research Question

Is there a relationship between traffic density and accident severity score?

Hypotheses

- H0: ρ = 0 (no linear correlation)
- H1: ρ ≠ 0 (two-sided test)
- Significance level α = 0.05
- Severity coded as Low = 1, Medium = 2, High = 3

Formula

$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \sqrt{(\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2)}$

$t = r\sqrt{(n - 2)} / \sqrt{(1 - r^2)}, df = n - 2$

Data Used (read from reference figure)

Quantity	Value
Sample size n	100
Pearson r (from figure)	0.72

Calculation

- $t = 0.72 \times \sqrt{98} / \sqrt{(1 - 0.72^2)} \approx 10.27$, df = 98

- Because severity is ordinal, Spearman's rank correlation is also reported as a robustness check.

Result

Quantity	Value
Pearson r	0.72
t statistic	10.27
df	98
Two-sided p-value	< 0.001

Conclusion

The p-value is far below 0.05, so we reject H0. There is a strong, statistically significant positive relationship between traffic density and accident severity score: higher density tends to go with higher severity.

Interpretation Notes

- r > 0 indicates a positive relationship; r ≈ 0.72 indicates a strong one.
- The scatter plot shows three horizontal bands because severity is a 3-level ordinal variable. Pearson correlation is reasonable for exploration, but Spearman's ρ (cor.test(..., method = "spearman") in R) is more appropriate and should be reported alongside it.
- Correlation does not establish causation.

Graph

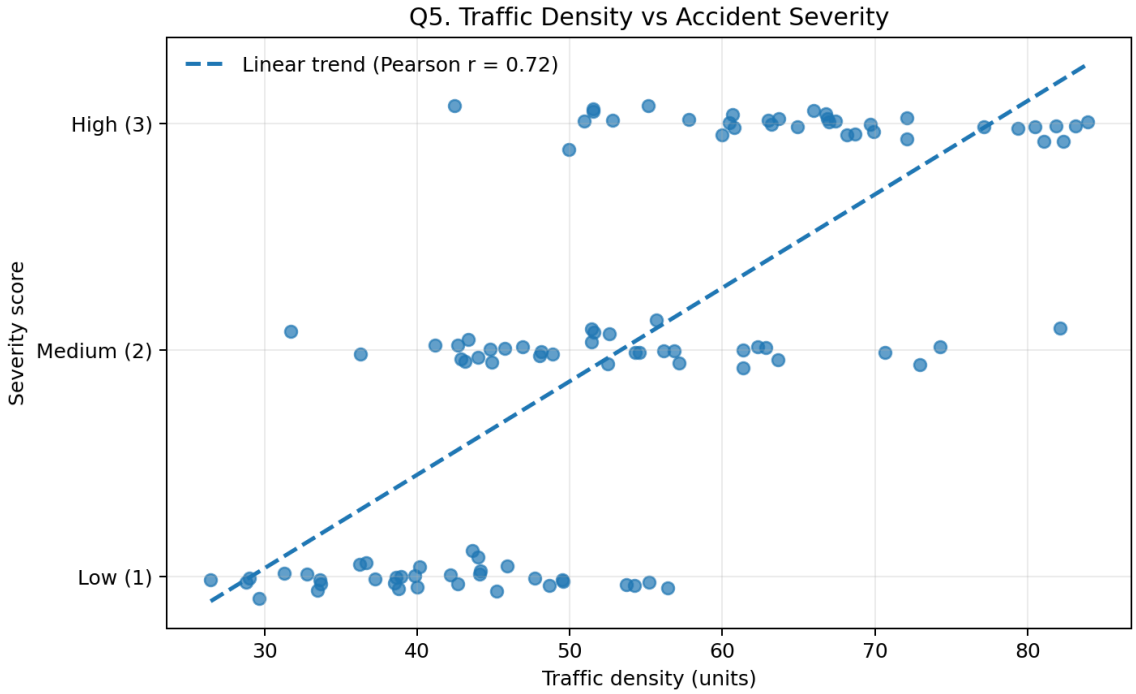


Figure 5. Reference graph for Question 5.

Note: the counts and values in this document were read from the reference figure. Re-run the test on the original project dataset in R (e.g. chisq.test, t.test, aov, cor.test) to confirm the final numbers before submission.